

YEAR 9

MATHS & SCIENCE

Completing Key Stage 3 and building the bridge into GCSE mathematics and science

Government-source edition • Department for Education / GOV.UK only

Prepared for families supporting pupils at the KS3-to-KS4 transition.

1. How to use this guide

Year 9 is a transition year, but there is no separate statutory DfE Year 9 syllabus. Government specifies what pupils should know by the end of Key Stage 3 and what must be taught at Key Stage 4. Schools decide when Year 9 ends KS3 content and when GCSE material begins.

The crucial point

Use Year 9 to close KS3 gaps before GCSE complexity compounds them. The right question is not “Have we started GCSE early?” but “Is the KS3 foundation secure enough for GCSE depth?”

2. Mathematics - the KS3 endpoint

DOMAIN	WHAT SHOULD BE SECURE BY THE END OF KS3
NUMBER	Integers, decimals and fractions; positive and negative numbers; factors, multiples and primes; four operations; powers, roots and reciprocals; exact calculation; rounding and estimation; fractions, decimals and percentages; standard form.
ALGEBRA	Notation and vocabulary; substitution; simplifying and manipulating expressions; identities and formulae; linear equations; sequences; coordinates; linear and quadratic graphs; interpreting gradients and intercepts; modelling.
RATIO & PROPORTION	Unit conversions; scale factors and scale diagrams; fractions as operators; direct and inverse proportion; percentage change; compound units such as speed, unit pricing and density.
GEOMETRY & MEASURES	Perimeter, area and volume; circles and composite

shapes; constructions; angles; properties of shapes; congruence and similarity; transformations; coordinates; Pythagoras and basic trigonometric relationships in appropriate contexts.

PROBABILITY	Probability scale; experimental and theoretical probability; mutually exclusive outcomes; sample spaces; systematic listing; interpreting simple combined events.
STATISTICS	Sampling; tables and charts; averages and range; comparing distributions; scatter graphs; correlation; interpreting and critiquing data presentations.

3. The Year 9 mathematics progression lens

MOVE FROM	MOVE TOWARDS
PROCEDURE	Selecting methods independently and explaining why they work.
SINGLE-STEP ALGEBRA	Multi-step manipulation, equations, graphs and modelling.
PERCENTAGES AS A TOPIC	Ratio, rates, proportion and percentage change used across contexts.
SHAPE FACTS	Geometric reasoning, similarity, constructions, Pythagoras and links between representations.
READING DATA	Comparing distributions, judging evidence and communicating conclusions.
FAMILIAR QUESTIONS	Unfamiliar, multi-step problems requiring perseverance and checking.

4. Science - the KS3 endpoint

DOMAIN	WHAT SHOULD BE SECURE BY THE END OF KS3
BIOLOGY	Cells and organisation; skeletal and muscular systems; nutrition and digestion; gas exchange; reproduction; health; photosynthesis; cellular respiration; relationships in ecosystems; inheritance, chromosomes, DNA and genes.
CHEMISTRY	Particle model; atoms, elements and compounds; pure

and impure substances; chemical reactions; energetics; periodic table; materials; Earth and atmosphere.

PHYSICS

Energy; motion and forces; waves and sound; light; electricity and magnetism; matter; space physics.

WORKING SCIENTIFICALLY

Scientific attitudes; experimental skills and investigations; analysis and evaluation; measurement, SI units, equations, graphs, uncertainty and communication of evidence.

5. The Year 9 science progression lens

MOVE FROM	MOVE TOWARDS
NAMING FACTS	Explaining observations with models and mechanisms.
FOLLOWING PRACTICALS	Planning, controlling variables and evaluating limitations.
READING GRAPHS	Using trends, gradients, equations and anomalies as evidence.
TOPIC ISLANDS	Connecting energy, particles, cells, forces, reactions and ecosystems.
WORDS ONLY	Using precise scientific vocabulary, symbols, equations and SI units.
RECALL	Applying knowledge to unfamiliar situations and judging evidence.

6. The GCSE bridge - what changes at Key Stage 4

- Mathematics becomes more cumulative: earlier number and algebra weaknesses interfere with later topics quickly.
- Science is taught in progressively greater depth; GCSE content deliberately builds on KS3 rather than simply repeating it.
- Assessment increasingly rewards application, reasoning, analysis and practical understanding - not recall alone.
- Schools may begin GCSE courses during Year 9, but their timing is a school curriculum decision rather than a national Year 9 mandate.

7. Parent readiness check

CHECK	GREEN SIGNAL	ACTION IF NOT SECURE
Maths fluency	Fractions, percentages, negatives, powers and algebraic manipulation are dependable.	Repair prerequisite gaps with short, frequent retrieval and mixed practice.
Maths reasoning	Student can explain, justify and select methods.	Ask “why this method?” and use unfamiliar problems.
Science knowledge	Core KS3 models and vocabulary can be retrieved without heavy prompting.	Build retrieval around cells, particles, energy, forces, reactions and ecosystems.
Science enquiry	Student can identify variables, interpret graphs and evaluate evidence.	Use practical write-ups and data questions, not only notes.
School alignment	Family knows the school’s Year 9/GCSE start point and pathway.	Obtain the published curriculum map and exam-board/pathway information when available.

8. What to obtain from your child’s school

- Year 9 maths and science curriculum maps by term.
- The point at which GCSE teaching formally begins.
- Whether science is taught as Combined Science or separate Biology, Chemistry and Physics.
- The GCSE awarding organisation/specification when confirmed.
- Tiering or set decisions, assessment dates and intervention thresholds.

Official government sources used

The curriculum content in this guide is derived from Department for Education / GOV.UK statutory programmes of study and GCSE subject-content publications. The year-specific sequencing lens is labelled as an Inspire support framework because government documents generally specify content by key stage or qualification, not by individual secondary-school year.

Department for Education. National curriculum in England: mathematics programme of study - key stage 3.

Ref: DFE-00179-2013

Published 11 September 2013; GOV.UK page updated 28 September 2021.

<https://www.gov.uk/government/publications/national-curriculum-in-england-mathematics-programmes-of-study>

Department for Education. National curriculum in England: science programme of study - key stage 3.

Ref: DFE-00185-2013

Published 11 September 2013; GOV.UK page updated 6 May 2015.

<https://www.gov.uk/government/publications/national-curriculum-in-england-science-programmes-of-study>

Department for Education. National curriculum in England: mathematics programme of study - key stage 4.

Ref: DFE-00496-2014

Published within the national curriculum mathematics programmes of study.

<https://www.gov.uk/government/publications/national-curriculum-in-england-mathematics-programmes-of-study>

Department for Education. National curriculum in England: science programme of study - key stage 4.

Ref: DFE-00677-2014

Published 2 December 2014 within the national curriculum science programmes of study.

<https://www.gov.uk/government/publications/national-curriculum-in-england-science-programmes-of-study>

Department for Education. GCSE mathematics: subject content and assessment objectives.

Ref: DFE-00233-2013

Published 1 November 2013.

<https://www.gov.uk/government/publications/gcse-mathematics-subject-content-and-assessment-objectives>

Department for Education. GCSE subject content for combined science.

Ref: DFE-00351-2014

Published 9 April 2014; last updated 3 July 2015.

<https://www.gov.uk/government/publications/gcse-combined-science>

Department for Education. GCSE subject content for biology, chemistry and physics.

Ref: DFE-00352-2014

Published 9 April 2014; last updated 3 July 2015.

<https://www.gov.uk/government/publications/gcse-single-science>